Modeling Shoulder Load Across Eras of the Royal Challengers Bengaluru: An Estimate-Based Biomechanics Framework for Batters, Wicketkeepers and Bowlers

Nikhil Chandrashaker, California Lutheran University. Working paper, November 2026.

Abstract

We build a transparent, estimate-based model of shoulder loading for 41 Royal Challengers Bengaluru (RCB) players drawn from three groups: the IPL 2026 squad, the 2025 title squad, and a core group from the 2011-2017 Kohli, de Villiers and Gayle years. The group covers 11 batters, 5 wicketkeepers, 14 all-rounders and 11 bowlers. Measured joint kinematics and kinetics for IPL players are not public, so the model combines estimated release and bat speeds, action-class factors and assumed seasonal workloads into a relative Shoulder Load Index (SLI), propagates uncertainty with Monte Carlo sampling, and adds a recovery model for fatigue across a 14-match season. In the base case frontline fast bowlers carry the highest load (Hazlewood, Bhuvneshwar, Dayal), followed by Vinay Kumar, Watson, the wicketkeeper-batter Salt and Thushara. The results are hypotheses to test against tracking and medical data, not clinical findings.

1. Introduction

The shoulder is repeatedly loaded in cricket by fast bowling, hard flat throws, wicketkeeping gathers and dives, and powerful batting swings. Workload-injury research in cricket, such as Dennis et al. (2003) on bowling workload in elite fast bowlers and Gabbett (2016) on training load and injury risk, motivates modeling exposure rather than only peak intensity. Franchises hold proprietary tracking data, but outside analysts can still build explicit, falsifiable models from public information. This paper does that for RCB across eras and documents every assumption.

2. Data

Players come from three overlapping groups: the IPL 2026 squad (25 players), the 2025 title squad (19 players, 16 of them also in the 2026 squad), and a selected 2011-2017 core (14 players including Kohli). The selection is a judgment sample of well-known RCB players, not a complete list of everyone who played for the franchise. For each player the dataset records role, batting hand, bowling type, estimated release speed (kph), estimated bat speed at impact (kph), an action class, era tags and an estimate-confidence tag (H, M, L). Speeds are rounded estimates from broadcast speed-gun readings and role archetypes, and are less certain for retired players. Handedness or bowling type is marked unverified for five 2026 players. Seasonal workloads (balls bowled, balls faced, throws) are assumptions describing a typical RCB season for each player, including partial seasons for players who spent little time at the club. They are not verified career statistics.

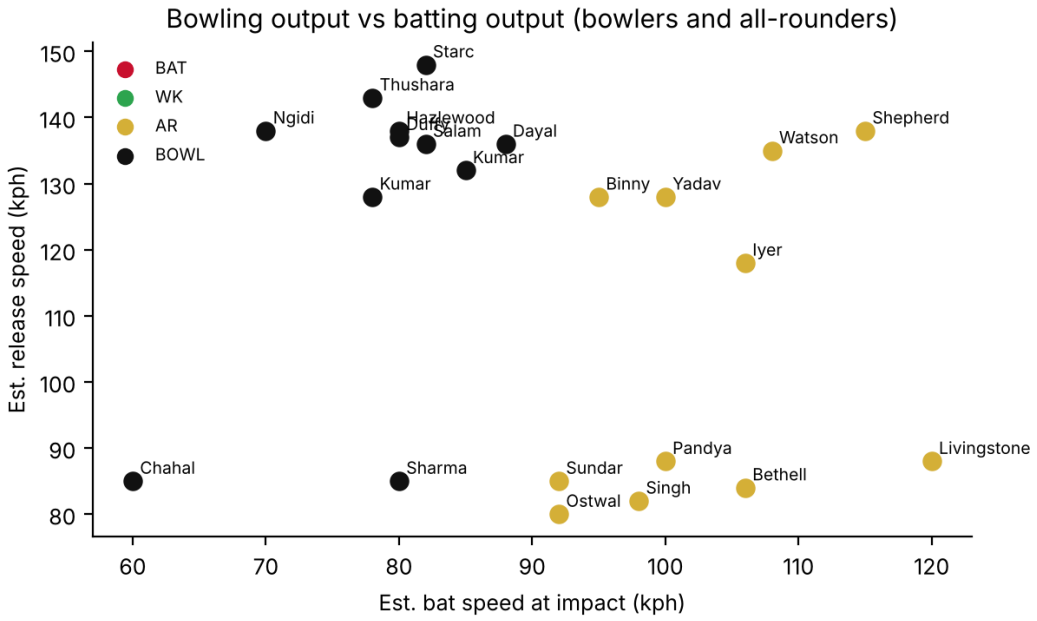


Figure 1. Estimated release speed against bat speed for bowlers and all-rounders across all three groups.

3. Methods

3.1 Impact model. Ball exit speed is $v_{out} = 0.95\ v_{bat} + 0.25\ v_{in}$, a calibrated linear form that folds in restitution, the bat-to-ball mass ratio and off-centre contact losses. Carry at a 30 degree launch is $0.72\ v^2 \sin(60\ deg)/g$, where 0.72 represents drag. The calibration puts elite hitters near 100 to 110 m (Figure 6). This supplies the bat-speed pathway, since swing speed scales rotational demand on the shoulder.

3.2 Shoulder Load Index. Three sources are summed. Bowling load = $(v_{rel}/143)^2 \times a \times N_{balls}/100$, where a is an action factor (low slinging 1.10, open-chested or round-arm 1.05, otherwise 1.00). Batting load = $0.35 \times (v_{bat}/118)^2 \times N_{faced}/100$, with the 0.35 weight reflecting the assumption that a swing stresses the shoulder less than a delivery. Fielding load = $p \times k \times N_{throws}/100$, with per-throw weight p = 0.50 for outfield throws and 0.25 for keepers and a dive-and-gather factor k = 1.5 for keepers. The sum is rescaled so the highest-load player in the 41-player set equals 100. Squaring speed follows kinetic-energy scaling and is a simplifying choice.

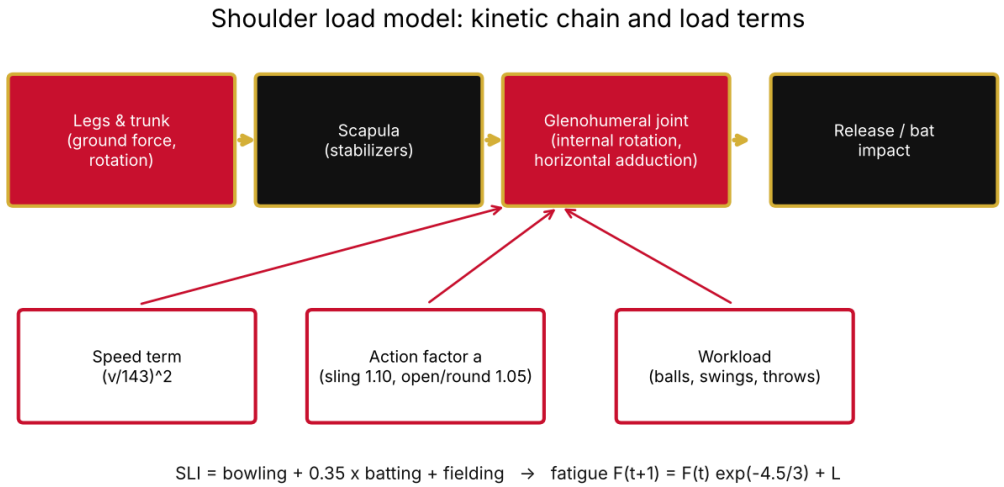


Figure 2. Model structure: the kinetic chain feeds the shoulder, and three terms set the load.

3.3 Fatigue accumulation. Season load is split evenly over 14 matches spaced 4.5 days apart. Fatigue evolves as F(t+1) = F(t) exp(-4.5/tau) + L, with tau = 3 days, so unrecovered load carries into the next match.

3.4 Uncertainty. For each player we ran 2,000 draws with speed scaled by a normal factor (mean 1, sd 3%) and workload scaled by a lognormal factor (sigma 0.2), reporting the 5th and 95th percentile of the SLI. Relative bands use the group median (56.0) and 80th percentile (66.2) of the base SLI. They are within-sample rankings, not clinical thresholds.

4. Results

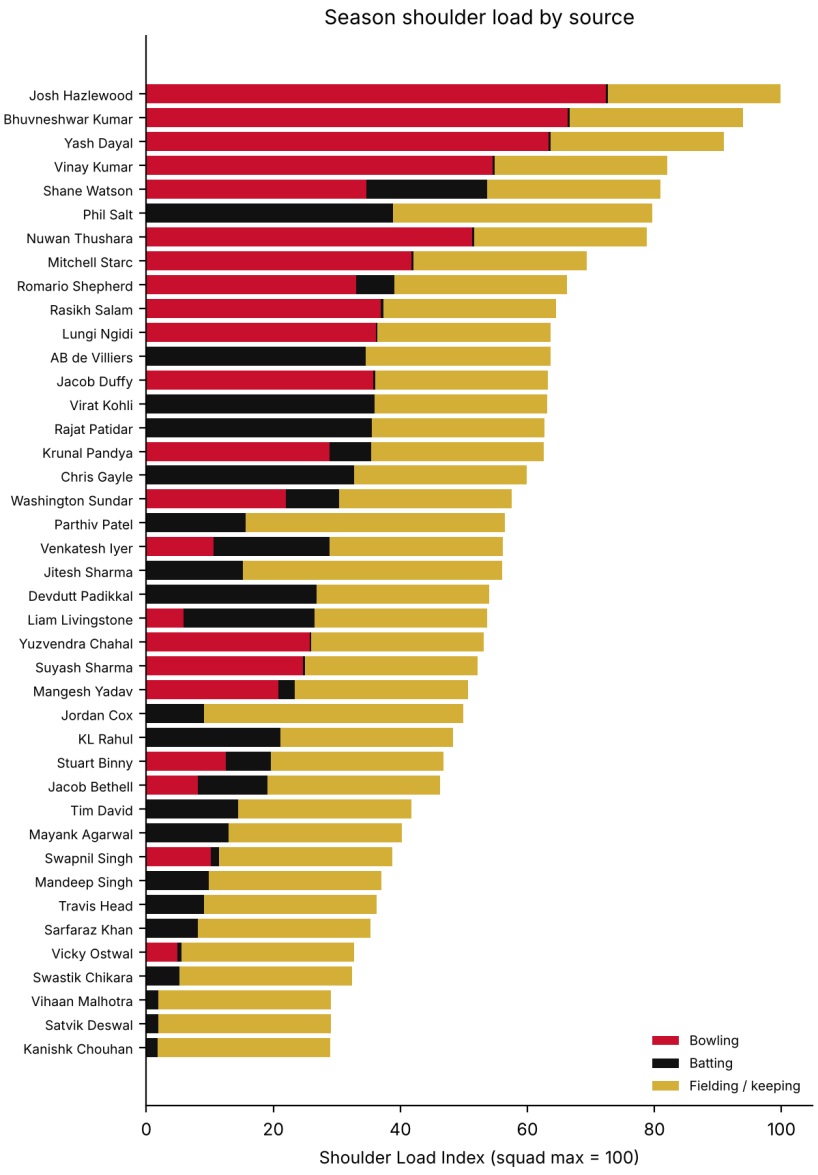


Figure 3. Season Shoulder Load Index by source for all 41 players.

Table 1. Twelve highest-load players in the base case.

Player	Role	Era group	SLI	Range	Band
Josh Hazlewood	BOWL	2026,2025	100.0	78-130	High
Bhuvneshwar Kumar	BOWL	2026,2025	93.9	74-119	High

Yash Dayal	BOWL	2026,2025	90.9	72-117	High
Vinay Kumar	BOWL	11-17	82.1	66-104	High
Shane Watson	AR	11-17	80.9	66-103	High
Phil Salt	WK	2026,2025	79.6	69-95	High
Nuwan Thushara	BOWL	2026,2025	78.9	63-100	High
Mitchell Starc	BOWL	11-17	69.2	57-87	High
Romario Shepherd	AR	2026,2025	66.2	55-82	High
Rasikh Salam	BOWL	2026,2025	64.5	54-79	Moderate
Lungi Ngidi	BOWL	2025	63.6	53-78	Moderate
AB de Villiers	WK	11-17	63.6	54-78	Moderate

Fast bowlers hold the top of the ranking. Hazlewood, Bhuvneshwar and Dayal lead, followed by the 2011-17 seamers Vinay Kumar and Watson, then Salt and Thushara. Starc ranks eighth on higher release speed but a smaller assumed workload. Salt is the highest-ranked non-bowler because he combines high bat speed, a heavy batting workload and keeping. Role means follow the same pattern: bowlers 73.8, keepers 61.1, all-rounders 48.5 and batters 46.5. Spinners and part-time bowlers rank lower because the squared speed term is small at 80 to 90 kph.

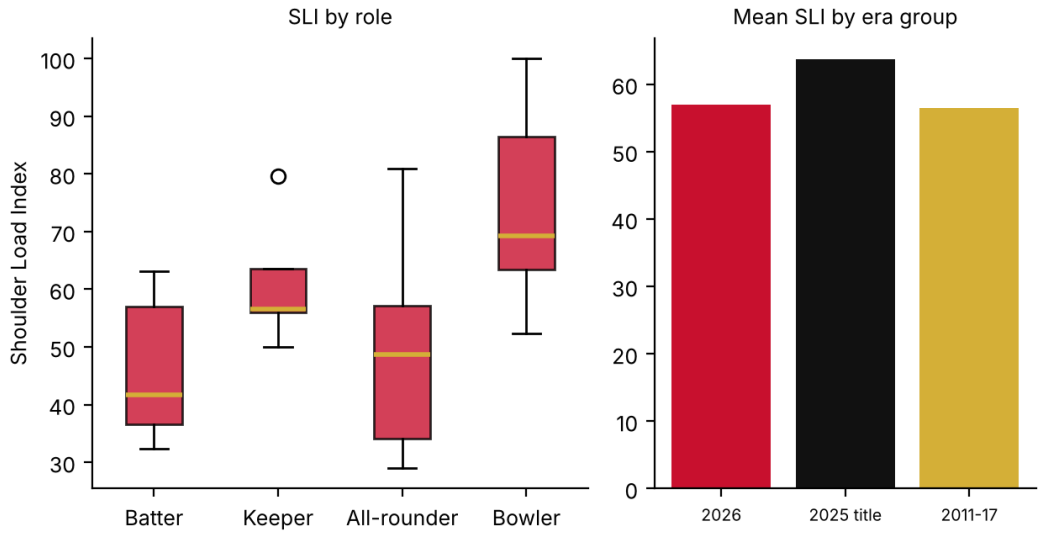


Figure 4. SLI by role (left) and mean SLI by era group (right).

The mean SLI is 56.9 for the 2026 squad, 63.6 for the 2025 title squad and 56.4 for the 2011-17 core. These differences should not be read as evidence that eras differ in shoulder demand. They mostly reflect which players are in each group and the workloads assumed for them, for example the title-squad group has a higher share of frontline bowlers. Mean estimated bat speed (about 98 kph) and mean estimated release speed (116 to 118 kph) are nearly identical across groups by construction of the estimates.

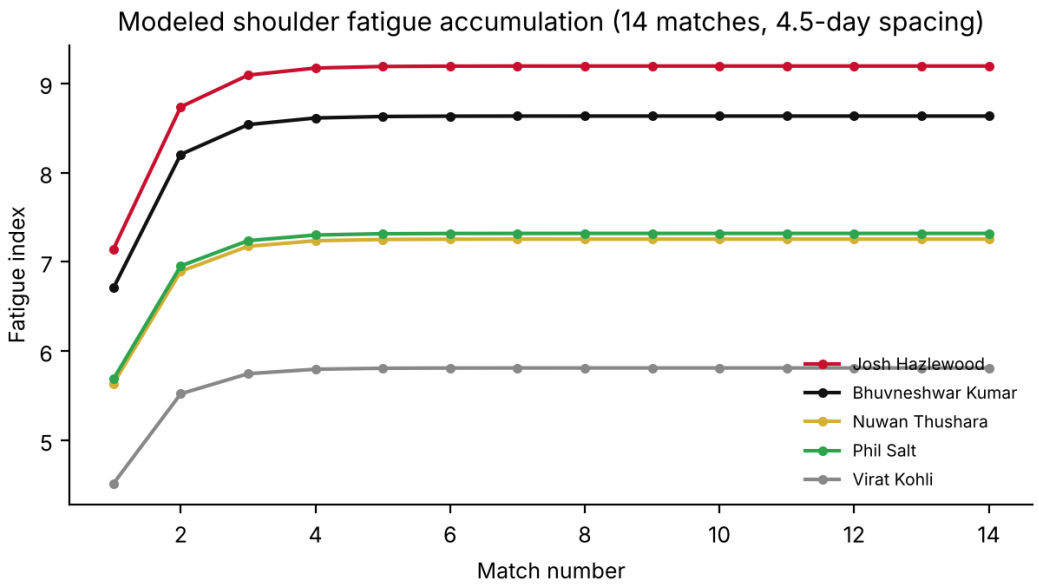


Figure 5. Modeled fatigue build-up over 14 matches for selected players.

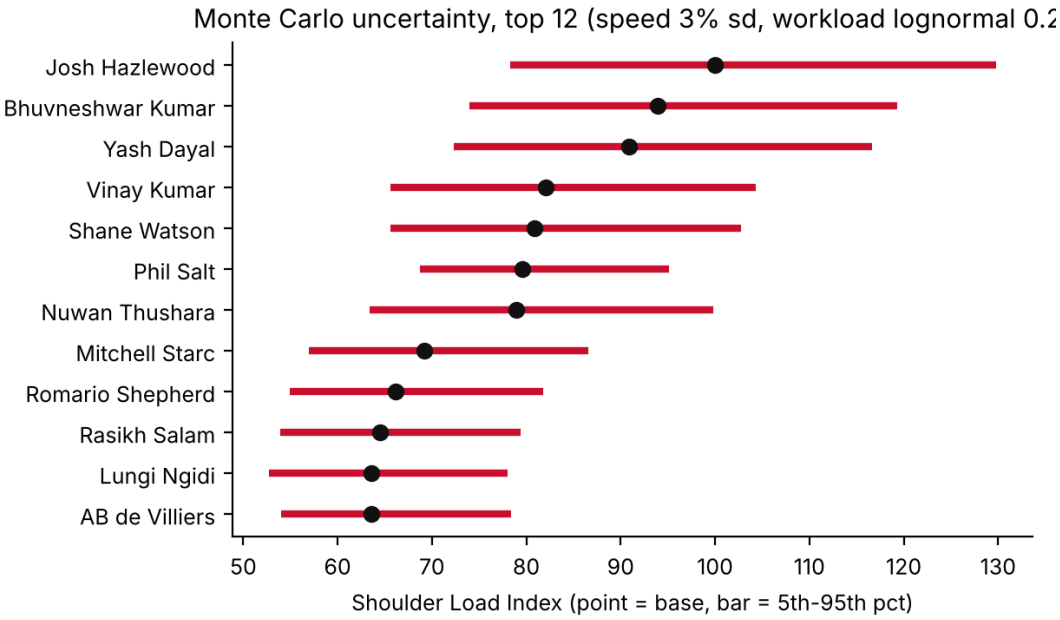


Figure 6. Monte Carlo 5th-95th percentile ranges for the 12 highest-load players.

Ranges overlap heavily between neighbouring players, so ordering inside a band is not robust. Only the gap between the fast-bowling group and the low-load group is stable across draws.

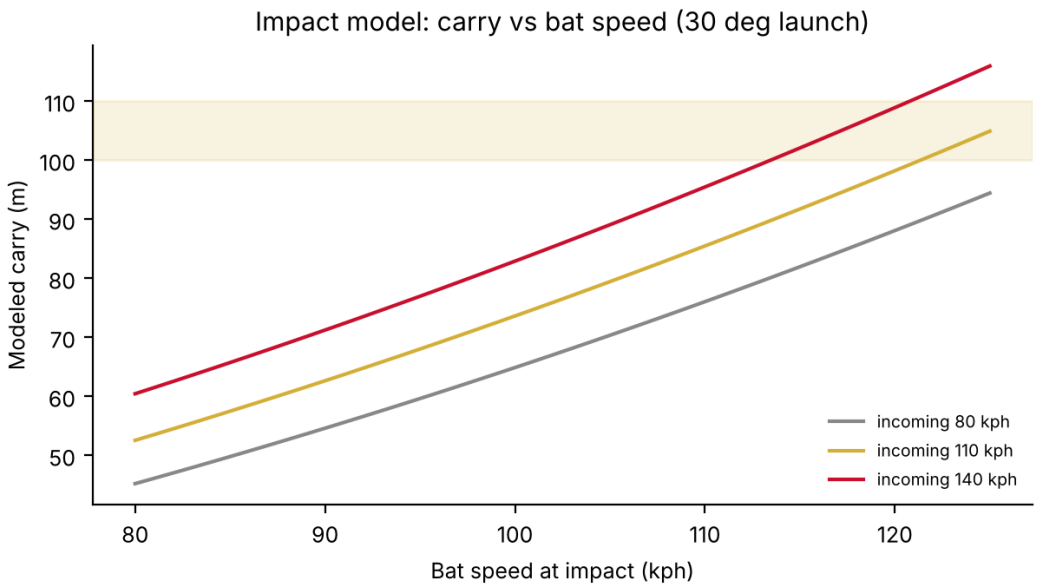


Figure 7. Impact-model carry as a function of bat speed for three incoming ball speeds. Shaded band is 100 to 110 m.

5. Discussion and Limitations

The SLI is only as good as its inputs, and the workload assumptions dominate the result: changing a player's assumed overs moves them more than any plausible error in release speed. Component weights are modeling choices, not fitted values. Speeds are estimates and are weaker for retired players, action factors are qualitative, the era groups are a judgment sample, and the model ignores run-up, fielding position, injury history, conditioning, travel and back-to-back fixtures. Shoulder load is not injury risk, and the model must not be used for medical or selection decisions.

6. Future Work

Replace assumed workloads with verified ball-by-ball and career data (balls bowled, balls faced, fielding events) for every player, extend to the full list of RCB players, fit weights against recorded shoulder-related absences where such data can be obtained, add per-delivery kinematics from markerless pose estimation of broadcast video, and test whether action class predicts economy, pace retention and injury timing.

References

Dennis R, Farhart P, Goumas C, Orchard J. Bowling workload and the risk of injury in elite cricket fast bowlers. J Sci Med Sport. 2003. (Verify citation details before submission.)
Gabbett TJ. The training-injury prevention paradox: should athletes be training smarter and harder? Br J Sports Med. 2016. (Verify citation details before submission.)
Squad data: IPL 2026 auction squad reports from public cricket news sites, December 2025. Past-era rosters are from general knowledge and should be verified.